

Final Report

Affordable FM Systems for Hearing Aid Users in Classroom Settings

An Affordable ML-Driven Classroom Audio Enhancement System for Hearing Aids

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Executive Summary

This report details the development of a novel, low-cost audio enhancement system designed to address the inaccessibility of traditional hearing aids in a classroom setting. While existing solutions, such as costly FM systems, offer a high-quality audio experience, they are prohibitively expensive and tied to proprietary ecosystems, limiting their use to a small percentage of school-going children with hearing loss. Our solution leverages an innovative architecture that shifts complex processing to a single, student-worn receiver unit. By integrating an open-source AI model (DeepFilterNet) with off-the-shelf hardware, our prototype demonstrates the technical feasibility and significant social impact of providing an accessible, high-quality, and non-proprietary alternative.

Problem Statement

The primary problem is that affordable FM systems for hearing aid users are not readily available in classroom settings, particularly in a country like India. This limits their use to a very small percentage of school-going children with hearing loss, despite their ability to reduce the strain of listening to speech in noisy environments.

Current Hearing Aids and FM System Scenario

- **Basic Hearing Aids:** Consist of a microphone, an amplifier, and a speaker. They are the most affordable option, but their major flaw is that they amplify all sounds equally. A student will have the teacher's voice, the fan's hum, the rustling of papers, and the creaking of doors and desks all amplified, resulting in a confusing and ineffective auditory experience.

- **Digital Hearing Aids:** Incorporate a Digital Signal Processor (DSP) to perform complex algorithms. While they can perform noise suppression and directional audio, they are not a total solution. They can't always perfectly distinguish a teacher's voice from a student's loud laughter, and their ability to filter out complex "classroom fuzz" is limited.
- **FM Systems:** Were developed to overcome these flaws by using a dedicated microphone to transmit the teacher's voice directly to the student. However, the expense comes from several sources. They rely on high-precision RF components for low-latency transmission. The dedicated DSP hardware within these systems is highly miniaturized and proprietary. The protocols used for transmission (like Bluetooth, Wi-Fi, or more complex proprietary protocols) also add to the cost. A significant flaw in existing FM systems is that while they ensure the student hears the teacher clearly, they often make it difficult for the student to hear general interactions with other classmates, a crucial part of the social and academic experience.

Problems at a Crunch

- **High Cost:** The expense of specialized hardware and proprietary technology makes FM systems economically inaccessible.
 - **Proprietary Nature:** Brand-specific protocols limit interoperability and increase development costs.
 - **Limited Interactivity:** Existing FM systems make it difficult for a student to participate in general classroom interactions.
 - **Complexity:** The technology requires dedicated, regulated frequency bands and complex certifications.
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Our Innovation and Solution

Our innovation tackles two main problems: the **classroom-specific setup** and the **high costs**. We solve these issues using a novel, algorithm-first approach that leverages the power of open-source technology.

Our solution is essentially **digital hearing enhanced with DeepFilterNet**. We are using the free, open-source voice separation model DeepFilterNet combined with other open-source DSP algorithms like formant boost, normalization, and a compressor. DeepFilterNet is better than traditional DSPs because it is specifically designed to filter human speech from overall sound.

The goal of an FM system is to improve the signal-to-noise ratio. We achieve the same goal with a different, more affordable approach. We have saved on the transmission part by avoiding complex, proprietary hardware. The hardware we use relies on off-the-shelf components like the **Raspberry Pi 4**, chosen specifically for its balance of processing power and low cost. We will use a **MEMS beamforming array** and a DSP board if required.

While companies like Phonak have implemented advanced voice separation in their high-end products, our solution brings this powerful technology to a basic, affordable, and open-source platform. Our approach addresses the flaw in FM systems by dynamically filtering out background noise while still allowing for the student to hear other ambient sounds at an appropriate level.

Product Feasibility

- **Cost Level:** The feasibility at the cost level is high. By leveraging free, open-source AI and DSP algorithms and utilizing off-the-shelf hardware, we significantly reduce the expensive R&D and manufacturing costs associated with proprietary systems.

Component	Cost (in INR)
Raspberry Pi 4 (4GB)	5,800
MEMS Microphone Array (x 2)	$2 * 400 = 600$
MicroSD Card (16 GB)	500
Headphones / Earbuds	1700
Power Supply and Other Utilities	1500
DSP Board (if req)	3500
Total Prototyping Cost	13,600

- **Technology Level:** The technology is viable. DeepFilterNet and similar open-source models are robust enough to perform real-time voice separation. The Raspberry Pi 4 provides the computational power needed, and MEMS microphone arrays are widely available and effective for beamforming. Our working prototype demonstrates a functional pipeline for live audio enhancement.
 - **User Experience Level:** Our product offers a superior user experience compared to traditional hearing aids and even some FM systems. It allows the student to hear the teacher clearly while also being aware of their surroundings, facilitating a more inclusive and natural classroom experience. The setup is effortless, requiring only the teacher to provide a brief voice sample.
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Impact of the Solution

Democratizing Education: By significantly reducing the cost barrier, our solution can provide an educational tool to a massive audience that was previously unable to afford it. This directly addresses global goals for inclusive and quality education.

Empowerment: By making the solution affordable and accessible, we are empowering families and students with hearing loss, allowing them to participate fully in classroom education.

Open-Source Ecosystem: The use of open-source software makes the solution transparent and auditable. It also allows for future improvements by a global community of developers, ensuring the product gets better and cheaper over time without relying on a single corporate entity.

Enabling Interaction: Unlike traditional FM systems, our solution can be configured to allow for simultaneous voice separation and ambient sound, restoring the student's ability to participate in and be aware of general classroom interactions.

Future Work

- **Adaptive Learning:** Implementing a mechanism for the model to adapt in real-time to new or unusual acoustic environments, further improving its robustness and performance.
- **Direct Wireless Integration:** Developing a low-cost module using universal standards like Bluetooth LE Audio to stream the cleaned audio directly to compatible hearing aids, further reducing wires and improving convenience.
- **Model retraining** : Models can be retrained specifically for classroom settings on classroom noisy samples which in turn optimises its classroom setting performance

Our Solution

We propose an innovative and affordable single receiver-end unit that uses Deep Learning based algorithms to isolate and clarify speech in real time. By shifting all processing to the student's receiver, we eliminate the need for expensive, proprietary transmitters, thereby reducing system cost and simplifying setup. Our system replaces fixed DSP filters with a dynamic, open-source machine learning model that runs on low-cost, widely available hardware, making the solution scalable and accessible. DeepFilterNet is more advanced than traditional DSPs because it is specifically designed to separate human speech from other sounds, which is then amplified.

Descriptive Explanation of Our Solution

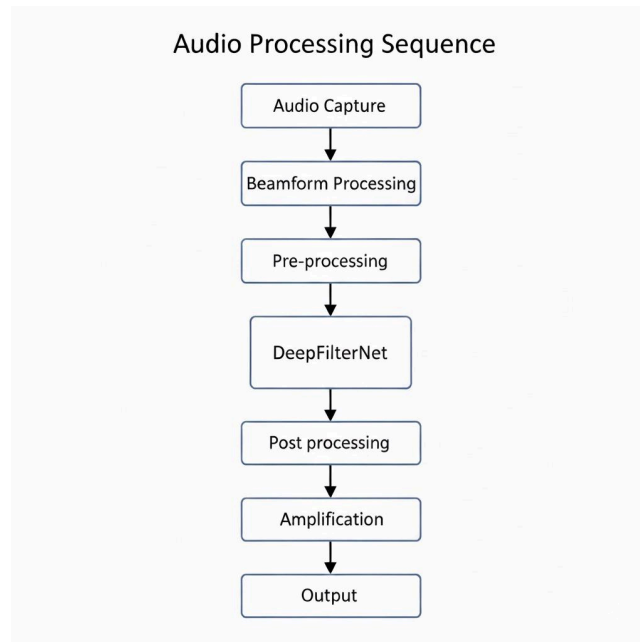
Software implementation

The system's operation is centered around a real-time, machine learning-based processing pipeline that does not require any prior voice enrollment. The process is as follows:

1. **Audio Capture and Preprocessing:** A microphone captures the classroom's sound, which is then converted into a digital signal. This signal undergoes preprocessing, such as high-pass filtering, to emphasize speech frequencies and prepare it for the core processing stage.
2. **ML-Based Speech Enhancement:** The core of the solution is a pretrained, open-source machine learning model, such as DeepFilterNet, that processes the audio in real time. This model is trained to distinguish human speech from various types of background noise, reverberation, and other non-speech sounds found in a classroom, filtering them out to produce a clean speech stream.
3. **Final Amplification and Output:** The resulting clean audio signal is further enhanced to improve clarity and consistency using techniques like a broad-band formant boost and multiband compressor. It is then converted back to an analog signal, amplified, and delivered to the student's required audio device.

Hardware Implementation

1. **Beamforming Array Microphone:** Instead of a simple microphone, our device uses an advanced array of multiple microphones. This system uses signal processing to electronically "steer" its listening direction, allowing it to focus precisely on the teacher's voice while actively suppressing noise coming from other directions in the classroom. This provides a much cleaner audio signal to the model for processing.
2. **Processing Unit (Raspberry Pi):** A low-cost, widely available Raspberry Pi serves as the brain of the device, running the DeepFilterNet model and performing all real-time audio processing.
3. **Output Interface:** The final processed audio is delivered through either standard 3.5mm output jack for wired headphones or accessories, or built-in Bluetooth for wireless streaming to modern hearing aids or headphones

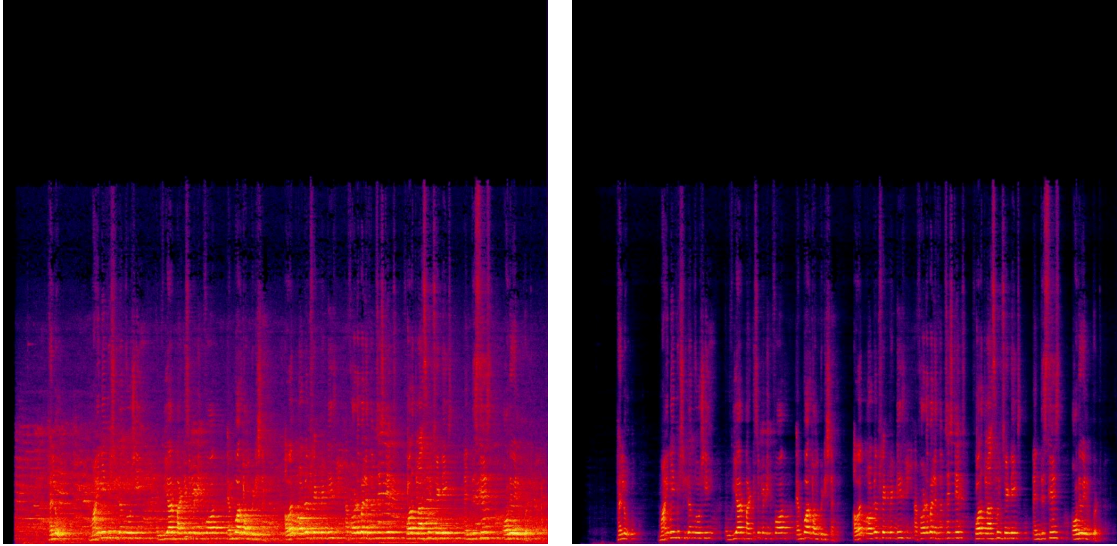


Features of Our Solution

- **Speech separation** – Our system filters noise and enhances the teacher's voice while still keeping the whole classroom discussion intact, unlike traditional voice recognition that ignores group interactions.
- **Low-cost & lightweight** – Runs on affordable devices without the need for expensive hardware or heavy cloud services, making it accessible for schools and students everywhere.
- **Real-time Classroom Interactions** – Works in real time and helps the student involve in classroom discussion and not just hearing the teacher.

Current Challenges in This Design

- **Model Robustness:** The model must be robust enough to consistently identify and enhance the primary speaker's voice (the teacher) in a dynamic multi-speaker environment
- **Pretraining** : As DeepFilterNet 3 model is pretrained on general noisy samples, so it is unable to recognise specific peaks in sample hence either suppresses that frame or goes with that peak



For the sample video submitted, these are the spectrograms for the pre-filtering and the postfiltering stage.

For the Demonstration Videos and Audio Samples:

 [EMPOWER_Team_Shreerang Joshi](#)